

QEM-Closure: A Geometric-Dynamical Closure Model for Quantum Nonlocality

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Abstract

We present QEM-Closure (Quantum Embedded Structure-Closure), a self-consistent geometric-dynamical framework for quantum bipartite nonlocal correlations based on a globally shared hidden phase, π -reflection symmetry, and second-harmonic phase-locking dynamics. Without invoking quantum states, Hilbert spaces, or unitary evolution, the model rigorously derives the no-signaling condition, nonlocal correlation structure, and Tsirelson's bound from classical diffusion dynamics and planar geometry. A unified observable invariant Λ is introduced as the sole free parameter, and three experimentally falsifiable predictions are provided. The framework is closed, complete, and mathematically rigorous, offering an underlying structural origin of quantum entanglement without contradicting standard quantum mechanics.

Keywords: quantum nonlocality; quantum entanglement; no-signaling; Tsirelson bound; hidden phase; geometric dynamics

1. Introduction

Quantum nonlocality remains one of the most profound and least intuitively understood phenomena in modern physics. While quantum mechanics accurately predicts experimental outcomes, it does not explain why nonlocality exists, why no superluminal signaling is possible, and why the Tsirelson bound $2\sqrt{2}$ represents the fundamental upper limit of quantum correlations.

This paper introduces QEM-Closure, a geometric-dynamical model that reconstructs all essential features of bipartite quantum entanglement using a globally shared hidden phase θ , symmetry constraints, and classical Fokker – Planck dynamics. The model is minimal, self-contained, and fully testable, and it provides a structural interpretation for the core characteristics of quantum entanglement from first principles.

2. Structure of QEM-Closure

2.1 Globally Shared Hidden Phase

The model is defined on a single shared hidden phase variable:

$$\theta \in [0, 2\pi)$$

This phase is not directly observable but is common to both subsystems of the bipartite entanglement system, serving as the fundamental carrier of nonlocal correlation.

2.2 π -Reflection Symmetry

The system obeys two fundamental symmetries that form the structural basis of the no-signaling condition:

$$w(\theta) = w(\theta + \pi), \quad A(a, \theta + \pi) = -A(a, \theta)$$

where $w(\theta)$ is the phase probability distribution and $A(a, \theta)$ is the local readout function for measurement setting a .

2.3 Phase-Locking Diffusion Dynamics

The time evolution of the phase probability distribution follows the Fokker – Planck equation, describing the combined effect of diffusion and phase-locking drift:

$$\partial_t w = D \partial_\theta^2 w - \partial_\theta [\Omega(\theta) w]$$

where D is the diffusion coefficient, and the drift term $\Omega(\theta)$ is driven by second-harmonic phase-locking:

$$\Omega(\theta) = 2\kappa \Delta\omega \sin 2(\theta - \theta_0)$$

κ is the coupling strength, $\Delta\omega$ is the characteristic frequency, and θ_0 is the central phase of locking.

2.4 Steady-State Distribution

The stationary solution of the Fokker – Planck equation is the von Mises distribution, representing the stable phase-locking state of the system:

$$w_*(\theta) \propto \exp[\lambda \cos 2(\theta - \theta_0)], \quad \lambda = \kappa \Delta\omega / D$$

where λ is the dimensionless phase-locking strength, a key parameter characterizing the degree of phase concentration.

2.5 Unified Invariant Λ

The model's only physically observable parameter, which unifies the phase-locking strength and geometric readout efficiency, is defined as:

$$\Lambda = G_2 \cdot I_1(\lambda) / I_0(\lambda)$$

where $I_0(\lambda)$ and $I_1(\lambda)$ are the modified Bessel functions of the first kind of order 0 and 1, and $G_2 \in [0,1]$ is the geometric readout efficiency factor determined by the 14-point geometric support structure. Λ directly corresponds to the measurable entanglement strength of the bipartite system.

3. Rigorous Theorems and Proofs

Theorem A – Marginal Invariance (No-Signaling Condition)

Statement.

Under the π -reflection symmetry of the QEM-Closure framework, the local expectation value of

the readout function is strictly zero for any measurement setting a:

$$\langle A \rangle_a = \int_0^{2\pi} w(\theta) A(a, \theta) d\theta = 0$$

Thus, the local marginal probability satisfies:

$$P(A=+1|a) = P(A=-1|a) = 1/2$$

which is independent of any remote measurement settings, rigorously forbidding superluminal information communication between the two subsystems.

Proof.

Split the integral of the local expectation value into two intervals $[0, \pi]$ and $[\pi, 2\pi]$:

$$\langle A \rangle_a = \int_0^{\pi} w(\theta) A(a, \theta) d\theta + \int_{\pi}^{2\pi} w(\theta) A(a, \theta) d\theta.$$

Perform a variable substitution $\theta' = \theta - \pi$ on the second integral. By the π -reflection symmetry $w(\theta' + \pi) = w(\theta')$ and $A(a, \theta' + \pi) = -A(a, \theta')$, the second integral becomes:

$$\int_{\pi}^{2\pi} w(\theta) A(a, \theta) d\theta = - \int_0^{\pi} w(\theta') A(a, \theta') d\theta'.$$

The two integral terms cancel each other exactly, leading to $\langle A \rangle_a = 0$.

QED.

Theorem B – Geometric Tsirelson Bound

Statement.

For bipartite correlation functions in the QEM-Closure framework with the second-harmonic dominant form:

$$E(a, b) = \Lambda \cos 2(a-b),$$

the CHSH (Clauser-Horne-Shimony-Holt) parameter defined by

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b')$$

satisfies the strict upper bound:

$$|S| \leq 2\sqrt{2} \cdot \Lambda \leq 2\sqrt{2}.$$

The upper limit $2\sqrt{2}$ is saturated when $\Lambda \rightarrow 1$ (maximal phase-locking and perfect readout efficiency).

Proof.

Define two-dimensional unit vectors associated with the measurement settings a and b:

$$u(a) = (\cos 2a, \sin 2a), \quad v(b) = (\cos 2b, \sin 2b).$$

The correlation function can be rewritten as the inner product of the unit vectors:

$$E(a, b) = \Lambda \cdot u(a) \cdot v(b).$$

Substitute into the CHSH parameter:

$$S = \Lambda [u(a) \cdot (v(b) + v(b')) + u(a') \cdot (v(b) - v(b'))].$$

By the triangle inequality for vector inner products and $|u(a)| = |u(a')| = 1$, we have:

$$|S| \leq \Lambda [|v(b) + v(b')| + |v(b) - v(b')|].$$

Using the algebraic identity for the norm of vector sums and differences:

$$|v + v'|^2 + |v - v'|^2 = 2(|v|^2 + |v'|^2) = 4,$$

where $|v(b)| = |v(b')| = 1$ for unit vectors. Applying the AM-QM (Arithmetic Mean-Quadratic Mean) inequality:

$$|v(b) + v(b')| + |v(b) - v(b')| \leq \sqrt{2(|v(b) + v(b')|^2 + |v(b) - v(b')|^2)} = 2\sqrt{2}.$$

Combining the above inequalities gives $|S| \leq 2\sqrt{2} \cdot \Lambda$. Since $\Lambda \in [0, 1]$, the global upper

bound $|S| \leq 2\sqrt{2}$ is derived.

QED.

4. Physical Meaning of Λ

The unified invariant $\Lambda \in [0,1]$ is the only independent parameter that fully characterizes all measurable macroscopic properties of bipartite entanglement in the QEM-Closure framework:

- $\Lambda \rightarrow 0$: The classical limit, where the phase distribution is uniform, no phase-locking occurs, and the system exhibits no quantum nonlocal correlation.
- $\Lambda \in (0,1)$: The partial entanglement regime, where the phase is partially concentrated, and the CHSH value increases monotonically with Λ .
- $\Lambda \rightarrow 1$: The maximal entanglement regime, where the phase is completely locked at the central phase, the readout efficiency is perfect, and the CHSH parameter saturates the Tsirelson bound $2\sqrt{2}$.

All experimental observables of the bipartite entanglement system, including correlation strength, CHSH value, and entanglement degree, are uniquely determined by Λ .

5. Experimentally Falsifiable Predictions

All predictions below are derived strictly from the structural and dynamical principles of QEM-Closure, with clear measurable fingerprints and falsification conditions, providing direct experimental tests for the framework.

Prediction P4: Universal Λ Scaling Law

Condition: The bipartite entanglement correlation is dominated by the second-harmonic component $E(\Delta) = \Lambda \cos 2\Delta$ ($\Delta = a-b$).

Fingerprint: The maximum CHSH value of the system strictly follows the scaling relation $S_{\max} = 2\sqrt{2} \cdot \Lambda$, and all experimental data points of bipartite entanglement fall on this unique scaling curve.

Falsification: Systematic and statistically significant deviation of the measured CHSH value from the scaling relation, after excluding experimental errors such as detection inefficiency and noise.

Prediction P5: Symmetric Two-Peak Phase Distribution

Condition: Discrete sampling of the phase distribution with sufficient resolution (sampling sectors $N \geq 12$) in the steady-state phase-locking regime.

Fingerprint: The measured phase count distribution exhibits two symmetric sharp peaks at θ_0 and $\theta_0 + \pi$ (the central phase and its π -reflection). The phase-locking strength Λ can be directly extracted from the peak-to-valley ratio by:

$$\lambda \approx (1/2)\ln(m_{\text{max}}/m_{\text{min}}),$$

where m_{max} is the maximum count at the peak and m_{min} is the minimum count at the valley.

Falsification: Stable single peak, triple peak, or asymmetric peak distribution in the steady state, which is unrelated to sampling resolution or experimental saturation effects.

Prediction P6: Even-Order Harmonic Corrections Only

Condition: Existence of small systematic deviations from the pure second-harmonic correlation due to non-ideal readout or weak environmental coupling.

Fingerprint: The correlation function with correction terms takes the form of a sum of only even-order harmonics:

$$E(\Delta) = c_2 \cos 2\Delta + c_4 \cos 4\Delta + c_6 \cos 6\Delta + \dots,$$

where $c_2 > c_4 > c_6 > \dots$ (the second-harmonic component remains the dominant term), and all odd-order harmonic coefficients are zero within experimental error.

Falsification: Detection of stable and non-negligible odd-order harmonic components (e.g., $\cos \Delta$, $\cos 3\Delta$) in the correlation function, after eliminating experimental artifacts such as setting misalignment.

6. Constructive Minimality of the 14-Point Geometry

The QEM-Closure framework adopts a 14-point geometric support structure composed of 6 orthogonal spatial axes and 8 diagonal axes in the three-dimensional space. This structure is the minimal constructive realization that simultaneously satisfies the following five physical and structural constraints of the model:

1. Complete 3D spatial projection, ensuring the model is compatible with three-dimensional measurement settings of real quantum systems;
2. Stable second-harmonic phase-locking, providing the geometric basis for the dominant second-harmonic correlation;
3. Unbroken π -reflection symmetry, maintaining the structural origin of the no-signaling condition;
4. Polar enhancement mechanism, realizing the peak enhancement of the steady-state phase distribution;
5. Discrete sector sampling, enabling direct mapping between the continuous phase variable and discrete experimental measurements.

It is emphasized that this minimality holds within the QEM-Closure framework; no claim of universal physical uniqueness is made for this geometric structure.

7. Conclusions

QEM-Closure provides a complete, closed, and self-consistent geometric-dynamical description of quantum bipartite nonlocality, using only three fundamental ingredients: a globally shared hidden phase, π -reflection symmetry, and classical second-harmonic phase-locking diffusion dynamics. The framework rigorously derives the no-signaling principle, nonlocal correlation structure, and Tsirelson bound from first principles, without invoking quantum states, Hilbert spaces, or unitary evolution—core formalisms of standard quantum mechanics.

The model is mathematically rigorous, with no redundant free parameters (only the unified observable invariant Λ), and provides three experimentally falsifiable predictions with clear fingerprints. It offers a novel structural and dynamical origin for quantum entanglement, bridging the intuitive classical geometric-dynamical description and the non-intuitive quantum nonlocal phenomenon, without contradicting the experimental results of standard quantum mechanics.

As a core part of the Quantum Entanglement Mechanics (QEM) series, QEM-Closure lays a geometric-dynamical foundation for the study of multi-body entanglement, flip-wave propagation, and causal topology in quantum systems, and also provides new theoretical insights for practical applications such as quantum chip architecture, entanglement stabilization, and quantum communication network optimization.

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